12/06/2022

FOURIER TRANSFORM

Non-periodic : $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) e^{iwt} \, dw$

$$\boxed{X(w) = \int_{-\infty}^{\infty} x(t) e^{-iwt} \, dt}$$

Periodic : $x(t) = x(t+T)$ periodic functios

$$X(w) = \left(\int_0^T x(t) e^{-iwt} \, dt \right) \underbrace{\left(\sum_{n=-\infty}^{\infty} e^{-inwT} \right)}_{\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(w - \frac{2\pi k}{T}\right)}$$

$$\boxed{X(w) = 2\pi \sum_{k=-\infty}^{\infty} a[k] \delta\left(w - \frac{2\pi k}{T}\right)}$$

- FT of $e^{-at} U(t)$ $\quad a \in R, \; a > 0$

$$\int_{-\infty}^{\infty} e^{-at} U(t) e^{-iwt} \, dt = \int_0^{\infty} e^{-at} e^{-iwt} \, dt$$

$$= \int_0^{\infty} e^{-(at+iwt)} \, dt$$

$$= \frac{e^{-(a+iw)t}}{-(a+iw)} \Big|_{t=0}^{\infty}$$

$|e^{i\theta}| = 1$ for all $\theta \in R$

$$= \frac{e^{-a\infty} e^{iw\infty}}{-(a+iw)} + \frac{1}{(a+iw)}$$

FOURIER TRANSFORM

- FT of $\delta(t)$

$$\int_{-\infty}^{\infty} \delta(t)\, e^{-i\omega t}\, dt = e^{-i\omega 0} = 1$$

- FT of $\delta(t-a)$

$$\int_{-\infty}^{\infty} \delta(t-a)\, e^{-i\omega t}\, dt = e^{-i\omega a}$$

$$S = \frac{a(r^n - 1)}{r-1}$$

- FT of $x_{(-T,T)}(t)$

$$\int_{-\infty}^{\infty} x_{(-T,T)}(t)\, e^{-i\omega t}\, dt = \int_{-T}^{T} e^{-i\omega t}\, dt$$

$$= \frac{e^{-i\omega t}}{-i\omega}\Big|_{t=-T}^{T}$$

$$= \frac{e^{i\omega T} - e^{-i\omega T}}{i\omega}$$

$$= 2\,\frac{\sin(\omega T)}{\omega}$$

- FT of 1 $\left(\lim_{T\to\infty} x_{(-T,T)}(t) = 1\right)$

$$\int_{-\infty}^{\infty} 1 \cdot e^{-i\omega t}\, dt = \lim_{T\to\infty} \int_{-T}^{T} e^{-i\omega t}\, dt$$

$$= \lim_{T\to\infty} 2\,\frac{\sin(\omega T)}{\omega}$$

$$= 2\pi\,\delta(\omega)$$

- FT of e^{iat} $a \in R$

$$\int_{-\infty}^{\infty} e^{iat} e^{-iwt}\, dt = \lim_{T\to\infty} \int_{-T}^{T} e^{\overbrace{it(a-w)}^{-i(w-a)t}}\, dt$$

$$= \lim_{T\to\infty} 2\frac{\sin((w-a)T)}{w-a}$$

$$= 2\pi\delta(w-a)$$

- FT of $U(t)$

$$\int_{-\infty}^{\infty} U(t)\, e^{-iwt}\, dt = \lim_{T\to\infty} \int_{0}^{T} e^{-iwt}\, dt$$

- Properties of Fourier Transforms

1) Linearity $f(t) \longrightarrow F(w)$

2) Time shift $x(t-a) = e^{-jwa} \cdot X(w)$

$$e^{-2(t-5)} U(t-5) \xrightarrow{FT} e^{-jw5} \cdot \frac{1}{jw+2}$$

3) $e^{jat} x(t) \xrightarrow{FT} X(w-a)$

4) $x(-t) \xrightarrow{FT} X(-w)$

5) $x(t) \longrightarrow X(w)$

$x(at) \rightarrow \frac{1}{|a|} X(\frac{w}{a})$

6) Convolution $x(t) \rightarrow X(w)$, $y(t) \rightarrow Y(w)$ $\quad x(t) * y(t) \rightarrow X(w) Y(w)$

7) Multiplication $x(t) y(t) \rightarrow \frac{1}{2\pi} X(w) * Y(w)$

8) $\frac{d}{dt} x(t) \longrightarrow jw X(w)$

9) $\int_{-\infty}^{t} x(s)\, ds \longrightarrow \frac{1}{jw} X(w) + \pi X(0) \delta(w)$

Discrete Fourier Series

$x[n] = x[n+N]$ N: period

$$x = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \qquad e^{\frac{2\pi k n}{N}} \qquad k = 0, 1, 2, \ldots, N-1$$

$$e_m = \begin{bmatrix} e^{i\frac{2\pi m \cdot 0}{N}} \\ e^{i\frac{2\pi m 1}{N}} \\ \vdots \\ e^{i\frac{2\pi m (N-1)}{N}} \end{bmatrix}, \quad 0 \le m < N$$

$$\begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} = a_0 \begin{bmatrix} e^{i\frac{2\pi 0 \cdot 0}{N}} \\ \\ \\ \\ \end{bmatrix} + a_1 \begin{bmatrix} e^{i\frac{2\pi 1 \cdot 0}{N}} \\ \\ \\ \\ \end{bmatrix} + \ldots + a_{N-1} \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

multiplication of F

$$|e_i|^2 = e_i \cdot e_i = N \qquad e_i \cdot e_j = N \delta_{ij}$$

$$a_k = \frac{1}{N} x e_k = \frac{1}{N} \sum_{<N>} x[n] e_k[n]^* = \frac{1}{N} \sum_{n=<N>} x[n] e^{-i\frac{2\pi}{N} kn}$$

Discrete Fourier Transform

$$x[n] = \sum_{k=<N>} a[k]\, e^{ik\frac{2\pi}{N}n} \qquad a[k] = \frac{1}{N}\sum_{n=<N>} x[n]\, e^{-ik\frac{2\pi}{N}n}$$

$$X(\Omega_0 k) = \sum_{n=<N>} x[n]\, e^{-ik\Omega_0 n}$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-i\Omega n} \qquad \text{continuous, period} = 2\pi$$

$$X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta\left(\Omega - \frac{2\pi k}{N}\right)$$

- Find the DFT of Kronecker delta $\delta[n]$

$$DFT(\delta[n]) = \sum_{n=-\infty}^{\infty} \delta[n]\, e^{-i\Omega n} \quad) ?$$

$$= e^{-i\Omega 0} = 1$$

- DFT of $x_{[-N,N]}[n]$ $N \in \mathbb{N}$

$$DFT(x_{[-N,N]}[n]) = \sum_{n=-\infty}^{\infty} x(-N, N)[n]\, e^{-i\Omega n}$$

$$= \sum_{n=-N}^{N} e^{-i\Omega n}$$

$$= e^{-i\Omega N} \sum_{n=0}^{2N} e^{i\Omega n}$$

$$= e^{-i\Omega N}\, \frac{1 - e^{i\Omega(2N+1)}}{1 - e^{i\Omega}} = e^{-i\Omega N}\, \frac{e^{i\Omega(N+\frac{1}{2})}\left(e^{-i\Omega(N+\frac{1}{2})} - e^{i\Omega(N+\frac{1}{2})}\right)}{e^{i\frac{\Omega}{2}}\left(e^{-i\frac{\Omega}{2}} - e^{i\frac{\Omega}{2}}\right)}$$

$$= \frac{e^{-i\Omega(N+\frac{1}{2})} - e^{i\Omega(N+\frac{1}{2})}}{e^{-i\frac{\Omega}{2}} - e^{i\frac{\Omega}{2}}} = \frac{\sin(\Omega(N+\frac{1}{2}))}{\sin(\frac{\Omega}{2})}$$

- DFT of 1

$$DFT(1) = DFT\left(\lim_{N\to\infty} x_{(-N,N)}[n]\right)$$

$$= \lim_{N\to\infty} \frac{\sin\left(\Omega\left(N+\frac{1}{2}\right)\right)}{\sin\left(\frac{\Omega}{2}\right)}$$

- DFT of $U[n]$

$$DFT(U[n]) = \sum_{n=-\infty}^{\infty} U[n] e^{-i\Omega n}$$

$$= \sum_{n=0}^{\infty} e^{-i\Omega n} = \lim_{N\to\infty} \frac{1 - e^{-i\Omega(N+1)}}{1 - e^{-i\Omega}}$$

$$= \frac{1}{1-e^{-i\Omega}} - \lim_{N\to\infty} \frac{e^{-i\Omega(N+1)}}{1-e^{-i\Omega}} \frac{e^{i\frac{\Omega}{2}}}{e^{i\frac{\Omega}{2}}}$$

$$\vdots$$

$$= \frac{1}{1-e^{-i\Omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\Omega - 2\pi k)$$

Angle Between

$$V_1 \cdot V_2 = |V_1||V_2| \cdot \cos\theta$$

$$\|v\|_2 = \sqrt{\sum_{i=1} a_i^2}$$

$$\|v\|_1 = \sum |a_i|$$

$$\theta = \arccos \frac{V_1 \cdot V_2}{\|V_1\| \|V_2\|}$$

$$V_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad V_2 = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad V_1 \cdot V_2 = 1\cdot 2 + 2\cdot 3 + 3\cdot(-1) = 5$$

$$\|V_1\| = 14 \quad \|V_2\| = 14 \quad \theta = \arccos \frac{5}{14}$$

- uncountably finite region

$$\|f(x)\|_2 = \sqrt{\int_{-\infty}^{\infty} f(x)^2 dx} \qquad f(x), g(x) \quad \frac{\int_{-\infty}^{\infty} f(x) g(x) dx}{\sqrt{f(x)^2 dx}\sqrt{g(x)^2 dx}}$$

dot product of two functions $\langle f(t), g(t) \rangle = \int_0^L f(t) g^*(t)\, dt$

$|f(t)| = \sqrt{\langle f(t), g(t) \rangle} = \sqrt{\int_0^L f(t) f^*(t)\, dt}$

angle between function $\cos\theta = \frac{\langle f(t), g(t) \rangle}{\|f(t)\| \, |g(t)|}$

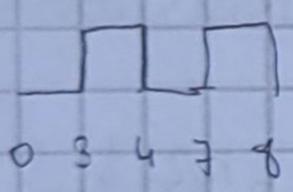
$e^{i \cdot \frac{2\pi}{L} \cdot kb}$ $\quad a_m = \sin\left(\frac{\pi}{5} m\right) \cdot \frac{1}{\pi m}$ $\quad i = e^{\frac{i\pi}{2}}$ $\quad e^{i\pi} + 1 = 0$

$\cos x = \mathrm{Re}(e^{ix}) = \frac{e^{ix} + e^{-ix}}{2}$ $\quad S = \frac{a(r^n - 1)}{r - 1}$

$\sin x = \mathrm{Im}(e^{ix}) = \frac{e^{ix} - e^{-ix}}{2i}$

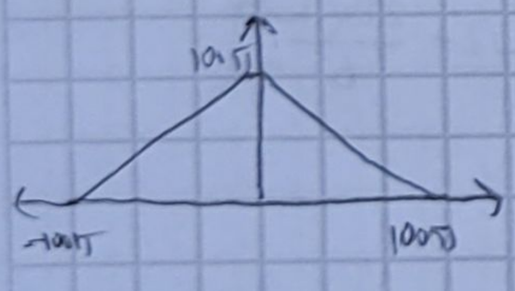
$\sin^n(ax+b)$ $\quad T = \frac{2\pi}{|a|}$ n = odd $\quad T = \frac{\pi}{|a|}$ n = even

$\cos^n(ax+b)$



$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} x_{[3,4]}(t - 4k)$

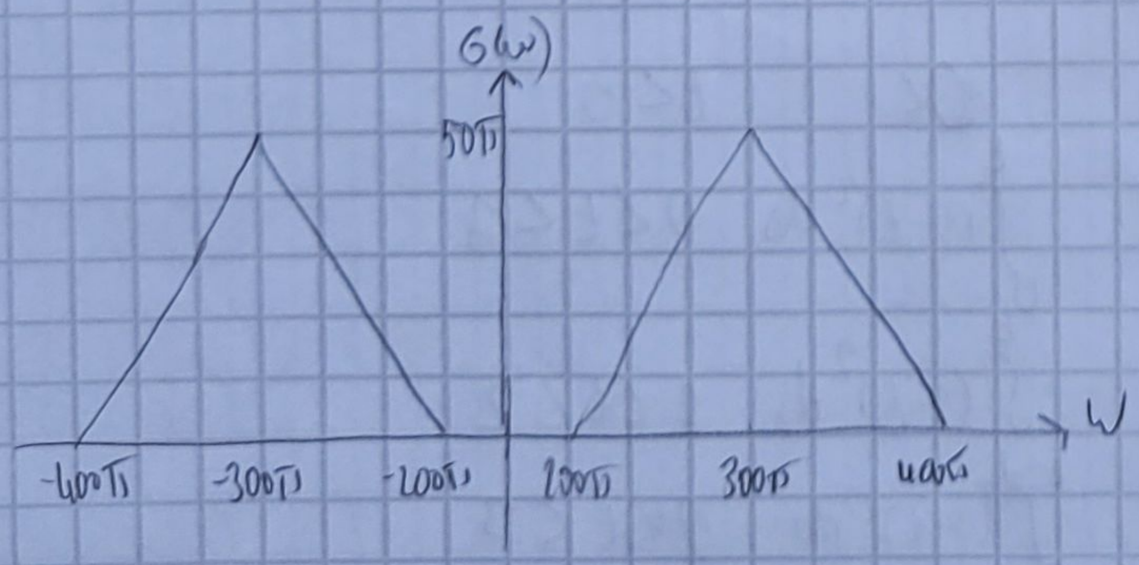
$F(W) = (w + 100\pi)\, x_{[-100\pi, 0]}(w) + (100\pi - w)\, x_{[0, 100\pi]}(w)$



$h(t) = \int_{-\infty}^{\infty} \cos(300\pi t)\, e^{-jwt}\, dt = \frac{1}{2} \int_{-\infty}^{\infty} \left(e^{300\pi j t} + e^{-300\pi j t}\right) e^{-jwt}\, dt$

$h(w) = \frac{1}{2} \delta(w - 300\pi) + \delta(w + 300\pi)$

$g(w) = f(w) \cdot g(w) = \frac{1}{2} \left[F(w - 300\pi) + F(w + 300\pi)\right]$



26/06/2022

- FT of $1+e^{-3|t|}(2U(t)-1)$

$$\underbrace{FT(1)}_{2\pi\delta(\omega)} + \underbrace{FT(e^{-3|t|}2U(t))}_{\frac{3\sqrt{2}}{\omega^2+9}} + \underbrace{FT(e^{-3|t|})}$$

$$\delta(t) = \lim_{\varepsilon\to 0}\frac{1}{A\varepsilon}f\left(\frac{t}{\varepsilon}\right) \qquad \sum_{n=0}^{k}\alpha^n = \frac{1-\alpha^{k+1}}{1-\alpha}$$

- FT of $\delta[n]$

- $X(\Omega) = \sum_{n=-\infty}^{\infty}\delta[n]e^{-j\Omega n} = 1$

- FT of $\delta[n-14]$

$$X(\Omega) = \sum_{n=-\infty}^{\infty}\delta[n-14]e^{-j\Omega n} = e^{-j\Omega 14}$$

- FT of $x_{(-N_1,N_1)}[n]$

$$\sum_{n=-\infty}^{\infty}\underbrace{x[n]}_{(-N_1,N_1)}e^{-j\Omega n} = \sum_{n=N_1}^{N_1}e^{-j\Omega n} = e^{-j\Omega N_1}\sum_{n=0}^{2N_1}e^{j\Omega n}$$

$$= e^{-j\Omega N_1}\cdot\frac{1-(e^{j\Omega})^{2N_1+1}}{1-e^{j\Omega}} = \frac{\sin[N_1+\frac{1}{2}]\Omega}{\sin[\frac{\Omega}{2}]}$$

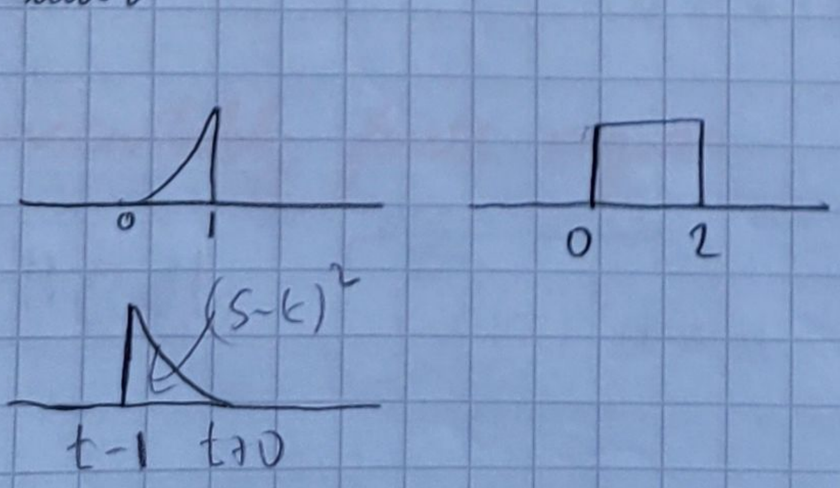
- $\sum_{k=-\infty}^{\infty}\delta(n-6k)\quad T=6$

$$\sum_{n=-\infty}^{\infty}\delta[n-6k]e^{-j\Omega n} = \sum_{k=-\infty}^{\infty}e^{-j\Omega 6k} \qquad X(\Omega) = \sum_{k=-\infty}^{\infty}\frac{1}{6}\delta\left[\Omega-\frac{2\pi 6k}{6}\right]$$

- 2 5 4 0 3 2

$$X(\Omega) = \sum_{n=-\infty}^{\infty}x[n]e^{-j\Omega n} \Rightarrow 2e^{j\Omega 2}+5e^{j\Omega}+4+3e^{-j\Omega 2}+2e^{-j\Omega 3}$$

- Convolve $t^2\chi_{(0,1)}(t)$ with $\chi_{(0,2)}(t)$ with



$$\begin{cases} 0 & t<0 \\ \int_0^t (s-t)^2\,ds & 0<t<1 \\ \int_0^t (s-t)^2\,ds & 1<t<2 \\ \int_{t-1}^{t} (s-t)^2\,ds & 2<t<3 \\ 0 & 3<t \end{cases}$$